

- 3 Fig. 3.1 shows a male red deer, *Cervus elaphus*. Red deer are herbivores, browsing on low vegetation in forests and on waste land.



Fig. 3.1

A history of wild red deer on the western European island of Ireland includes these facts:

- Red deer have lived in Ireland for at least 12 000 years.
- Originally red deer could cross from the neighbouring island of Great Britain to Ireland over a land connection.
- A rise in sea levels at the end of the last Ice Age removed this land connection, separating the red deer on the two islands.
- In the 1800s the number of red deer in Ireland decreased sharply after the main food crop for the human population failed for several consecutive years.
- In the 1900s this decrease in the number of red deer continued as large areas of waste land were drained for agriculture.
- By 1960 red deer were nearly extinct in Ireland, restricted to one population, **A**, of 60 individuals.
- Since then protection has allowed population **A** to increase to over 600 red deer.
- Several new red deer populations, **B**, **C** and **D**, have also been established in different parts of Ireland from individuals brought from Great Britain.

- (a) Use the information given to identify **two** causes of extinction that may have threatened the survival of red deer in Ireland.

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- (b) Describe how the level of molecular similarity between the red deer in population **A** and population **B** can be investigated.

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- (c) Explain why some red deer in population **A** show unique molecular features that are not found in any of the red deer of populations **B**, **C** and **D**.

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